delhi: A Declarative Language and Model Checker for Epistemic Reasoning over Plausibility Models

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Abstract

We present delhi, a declarative language and accompanying model checker for reasoning about the knowledge and beliefs of multiple agents, and about how those attitudes change when events occur that some agents witness and others do not. delhi implements mB^+, a plausibility-model semantics obtained by extending Buckingham's mB with the safe-belief and conditional-belief operators of Baltag and Smets, together with conditional effects and a systematic catalogue of derived attitudes. The distinguishing feature of the plausibility setting is that a single preorder per agent generates *both* knowledge and belief: knowledge quantifies over an agent's whole comparability class, belief over the maxima of its ordering. This yields a system in which an agent may be confidently and coherently wrong—about the world, and, more interestingly, about another agent's mind—and in which belief revision is a reordering rather than a deletion, so that a lie can be told and later corrected.

We give the syntax and semantics of the formula language, the surface language in which domains are written, and the action layer, in which the modeller declares *who observes what* and the event models are derived rather than hand-built. We formalise a query language that is exactly the formula language extended with a hole, and an enumeration procedure over modal literals that answers questions of the form "what does this agent believe that is false?" rather than requiring the analyst to guess the formula. We report the frame properties the implementation verifies ($S5$ for knowledge, $KD45$ for belief, factivity for safe belief), the bisimulation contraction that makes iterated update tractable, and measurements showing that contraction converts exponential growth into a fixed point on some domains but demonstrably not on all. We close by positioning mB^+ against DEL, $m\mathcal{A}^*$, EFP and PDKB, and by stating plainly which components are transcribed from the source literature and which are new.

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1 Introduction

Most formalisms for reasoning about action track *facts*: the coin is heads, the marble is in the basket, the robot is in the hall. A smaller family tracks *attitudes toward* facts—what each agent knows, what it believes, what it believes about what another agent believes—and it is in that second family that the phenomena of deception, coordination, teaching, and pretence live. All of them require an agent's model of another agent to be capable of going stale, or of being wrong.

The canonical illustration is a task from developmental psychology. Sally puts her marble in a basket and leaves the room; Anne moves it to a box; Sally returns. Where will Sally look? The answer is the basket, and producing it requires holding two incompatible pictures at once: where the marble is, and where Sally takes it to be. Children below roughly four years answer "the box" (Wimmer and Perner, 1983), and a representation that records only facts is in the same position—it has no machinery for a belief that is false.

delhi is a tool for building and interrogating such representations. A domain is written as a declarative file naming agents, propositions, what holds initially, what each agent believes initially, and a set of actions annotated with who observes them. From that file delhi constructs a plausibility model, derives event models for each action, and answers arbitrary queries in the modal language—including queries nested several levels deep.

1.1 Contributions

1. **A formalisation of mB$^+$** (Section 3), extending mB (Buckingham, 2021) with safe belief $\Box_i$ and conditional belief B_i^{ψ}, taken from Baltag and Smets (Baltag and Smets, 2008),

together with conditional ontic effects. We are explicit that the operators are not new; what is new is their presence in the object language of an action formalism, where they become queryable properties of a planning state.

2. **A surface language** (Section 4) in which initial states are given *declaratively*—as a list of facts and attitudes that the model must satisfy—rather than as a hand-drawn Kripke structure, together with non-recursive definitions, Horn rules over static constants, and state invariants.

3. **A query language and enumeration procedure** (Section 6). Queries are formulas of the full modal language extended with a hole _; enumeration ranges over modal literals to depth d, following the representation of Muise et al. (2015). Repeated holes are filled uniformly, which is what allows a single pattern to express "what does a believe that is false" as $B_{a}\_ \wedge \neg\_$.

4. **An implementation and its measurement** (Sections 7 and 8). We report the effect of bisimulation contraction on iterated update and show that, while it converts exponential growth into a fixed point on several standard domains, there are domains on which it demonstrably does not.

5. **An explicit accounting** (Section 9) of which parts of the system are transcribed from the literature and which are new, including three points at which the source documents disagree with each other and the resolution adopted.

1.2 Scope

delhi is a model checker, not a planner and not a theorem prover. It answers whether a formula holds in a given finite state, and computes the state resulting from a named action. It does not search for a sequence of actions achieving a goal; plan existence in the general dynamic-epistemic setting is undecidable (Bolander and Andersen, 2011), and the restrictions that recover decidability are orthogonal to the semantic questions treated here. Validity across all models is likewise out of scope: every judgement reported is a judgement about a particular pointed model.

2 Preliminaries

We assume finite sets $\mathcal{P}$ of propositional atoms and $\mathcal{G}$ of agents.

2.1 Plausibility models

The central object is a Kripke structure in which each agent's accessibility relation is a *preorder* rather than an equivalence, read as a comparative judgement of plausibility.

Definition 2.1 (Plausibility model)**.** A *plausibility model* is a triple $M = \langle W, R, V \rangle$ where

- W is a finite non-empty set of *worlds*;
- $R : W \times W \to (\mathcal{G} \to \mathbb{B})$, written uR_iv for $R(u, v)(i) = \text{true}$ and read *"agent i considers v at least as plausible as u"*;
- $V : \mathcal{P} \to 2^W$ is a valuation.

Each R_i is required to be *locally well-preordered*: a preorder (reflexive and transitive) whose restriction to any comparability class is connected. A *state* is a pointed model $\langle M, u \rangle$ with $u \in W$ the *actual* world.

Remark 2.2 (Direction of the ordering)**.** Plausibility *increases* along the arrow: uR_iv says v is the more credible of the two. This convention is worth stating explicitly because a model built with the relation inverted is still a well-formed model—it satisfies every frame condition of Definition 2.1—so no structural validation can detect the error. Every answer the system gives will simply be reversed.

Two relations are derived from R_i, and the whole of the semantics rests on the contrast between them.

Definition 2.3 (Comparability and maxima)**.** For a plausibility model M, agent i, world u and $C \subseteq W$:

$$u \sim_i v \quad \text{iff} \quad uR_iv \ \text{ or } \ vR_iu, \qquad \text{(comparability)} \tag{1}$$
$$\sim_i^u \ := \ \{\, v \mid u \sim_i v \,\}, \qquad \text{(the class of } u\text{)} \tag{2}$$
$$\to_i C \ := \ \{\, u \in C \mid u'R_iu \text{ for all } u' \in C \,\}, \qquad \text{(maxima of } C\text{)} \tag{3}$$
$$\to_i^u \ := \ \to_i \sim_i^u \,. \tag{4}$$

Proposition 2.4. *$\sim_i$ is an equivalence relation on W.*

Proof. Reflexivity and symmetry are immediate from the definition and the reflexivity of R_i. For transitivity, suppose $u \sim_i v$ and $v \sim_i w$. Local connectedness makes R_i total on the comparability class containing u and v, and v and w lie in that same class; hence u and w are comparable. □

Proposition 2.5 (Non-emptiness of maxima)**.** *If C is a non-empty finite subset of a single comparability class, then $\to_i C \neq \emptyset$.*

Proof. Local connectedness makes R_i restricted to that class a total preorder. A finite non-empty total preorder has maximal elements. □

Remark 2.6 (A precondition that must not be dropped)**.** Proposition 2.5 is stated in Buckingham (2021) without the single-class restriction, and in that generality it is false: if C contains worlds drawn from two distinct comparability classes, no element of either is R_i-related to any element of the other, so $\to_i C = \emptyset$. Both uses of $\to_i$ in mB^+ respect the restriction—$\to_i \sim_i^u$ takes a whole class, and the conditional-belief case of Definition 3.2 intersects a class with a denotation and treats the empty case explicitly—but an implementation that asserts non-emptiness unconditionally will fail on inputs the semantics permits.

2.2 Why one relation and not two

An alternative design gives each agent two relations, one equivalence for knowledge and one serial-transitive-Euclidean relation for belief, and constrains them to interact. The plausibility approach derives both from a single preorder, which has two consequences that matter for this paper.

First, the interaction axioms come for free rather than being imposed. Because $\to_i^u \subseteq \sim_i^u$ by construction, $K_i\varphi \to B_i\varphi$ is valid on every model, and no separate frame condition is required to secure it.

Second—and this is the point of the exercise—belief revision becomes a *reordering* rather than a deletion. Learning that ψ does not remove the $\neg\psi$-worlds from the model; it promotes the ψ-worlds above them. The agent's knowledge is therefore unchanged by an announcement, while its belief may be overturned, and because the demoted worlds are still present, a subsequent truthful announcement can promote them back. A semantics that deleted worlds would make the first lie irreversible.

3 The logic mB$^+$

3.1 Syntax

Definition 3.1 (Formula language)**.** The language $\mathcal{L}^{\mathcal{P}}_{\mathcal{G}}$ is generated by

$$\varphi \ ::= \ p \ | \ \neg\varphi \ | \ \varphi \wedge \varphi \ | \ K_i\varphi \ | \ B_i\varphi \ | \ C_g\varphi \ | \ B_i^{\psi}\varphi \ | \ \Box_i\varphi$$

with $p \in \mathcal{P}$, $i \in \mathcal{G}$, $g \subseteq \mathcal{G}$, and $\psi \in \mathcal{L}^{\mathcal{P}}_{\mathcal{G}}$. We write $\mathcal{L}^{\mathcal{P}}$ for the propositional fragment (the first three clauses). The connectives $\vee$, $\rightarrow$, $\top$ and $\bot$ are defined as usual.

The first six clauses are those of mB (Buckingham, 2021, Def. 1). The last two, conditional belief and safe belief, are the extension marked by the +; they are Baltag and Smets's operators (Baltag and Smets, 2006, 2008), and were already interpretable in mB's models, since those models *are* plausibility models. What mB does not do is lift them into its object language. mB$^+$ does.

3.2 Semantics

Definition 3.2 (Satisfaction)**.** For a state $\langle M, u\rangle$ with $M = \langle W, R, V\rangle$:

$$\langle M,u\rangle \models p \quad \text{iff} \quad u \in V(p) \tag{5}$$

$$\langle M,u\rangle \models \neg\varphi \quad \text{iff} \quad \langle M,u\rangle \not\models \varphi \tag{6}$$

$$\langle M,u\rangle \models \varphi \wedge \chi \quad \text{iff} \quad \langle M,u\rangle \models \varphi \text{ and } \langle M,u\rangle \models \chi \tag{7}$$

$$\langle M,u\rangle \models K_i\varphi \quad \text{iff} \quad \langle M,v\rangle \models \varphi \text{ for all } v \in \sim_i^u \tag{8}$$

$$\langle M,u\rangle \models B_i\varphi \quad \text{iff} \quad \langle M,v\rangle \models \varphi \text{ for all } v \in \rightarrow_i^u \tag{9}$$

$$\langle M,u\rangle \models \Box_i\varphi \quad \text{iff} \quad \langle M,v\rangle \models \varphi \text{ for all } v \text{ with } uR_iv \tag{10}$$

$$\langle M,u\rangle \models B_i^{\psi}\varphi \quad \text{iff} \quad \langle M,v\rangle \models \varphi \text{ for all } v \in \rightarrow_i (\sim_i^u \cap [\![\psi]\!]) \tag{11}$$

$$\langle M,u\rangle \models C_g\varphi \quad \text{iff} \quad \langle M,v\rangle \models \varphi \text{ for all } v \text{ with } u \left(\textstyle\bigcup_{i\in g} \sim_i\right)^* v \tag{12}$$

where $[\![\psi]\!] = \{v \mid \langle M,v\rangle \models \psi\}$ and $(\cdot)^*$ denotes reflexive-transitive closure. The conditional-belief clause is vacuously true when $\sim_i^u \cap [\![\psi]\!] = \emptyset$.

Three points deserve comment.

Common knowledge, not common belief. C_g is built on $\sim_i$, the comparability relation, and is therefore common *knowledge*. Some presentations use the same symbol for the closure over R_i, which yields common *belief*; the two are not interchangeable, and delhi implements the knowledge reading throughout.

B_i is a special case of B_i^{ψ}. Taking $\psi = \top$ gives $\sim_i^u \cap W = \sim_i^u$, so $B_i^{\top}\varphi \equiv B_i\varphi$. The implementation tests this identity rather than assuming it.

Safe belief is anchored to the actual world. K_i quantifies over a fixed set (the class), B_i over another fixed set (the maxima). $\Box_i$ quantifies over $\{v : uR_iv\}$, which depends on *where u sits* in the ordering. This gives $\Box_i$ a character quite unlike the other two, developed in Section 3.5.

3.3 The strength hierarchy

Theorem 3.3 (Hierarchy)**.** *For every model, agent and formula,*

$$K_i\varphi \ \rightarrow \ \Box_i\varphi \ \rightarrow \ B_i\varphi,$$

and neither implication is reversible.

Surface	Notation	Definition	Reading
`K'[a]`φ	$\hat{K}_a\varphi$	$\neg K_a\neg\varphi$	considers possible
`B'[a]`φ	$\hat{B}_a\varphi$	$\neg B_a\neg\varphi$	does not rule out
`S'[a]`φ	$\hat{S}_a\varphi$	$\neg\Box_a\neg\varphi$	safe-belief dual
`Kw[a]`φ	$Kw_a\varphi$	$K_a\varphi \vee K_a\neg\varphi$	knows whether
`Bw[a]`φ	$Bw_a\varphi$	$B_a\varphi \vee B_a\neg\varphi$	believes whether
`?[a]`φ	$?_a\varphi$	$\neg Kw_a\varphi$	ignorant whether
`??[a]`φ	$¿_a\varphi$	$\neg Bw_a\varphi$	suspends judgement
`C[*]`φ	$C_{\mathcal{G}}\varphi$	$C_g\varphi,\ g = \mathcal{G}$	common knowledge
`K[a,b]`φ	—	$K_a\varphi \wedge K_b\varphi$	agent-list distribution

Table 1: Derived attitudes. Agent lists distribute over K, B, $\Box$, Kw, Bw, ? and ??. Each is a Boolean combination of the primitives, so the evaluator implements eight cases and no more.

Proof. The three operators quantify over $\sim_i^u \supseteq \{v : uR_iv\} \supseteq \to_i^u$. The first inclusion holds because uR_iv implies $u \sim_i v$. The second holds because a maximal element of the class is R_i-above every element of it, u included. A larger quantification domain yields a stronger claim, giving both implications. Failure of the converses is witnessed in Example 3.6. $\Box$

Theorem 3.4 (Frame properties)**.** *On every plausibility model:*

1. *K_i satisfies T, 4 and 5, hence is an $S5$ modality;*
2. *B_i satisfies D, 4 and 5, hence is a $KD45$ modality, and does* not *satisfy T;*
3. *$\Box_i$ is factive: $\Box_i\varphi \to \varphi$;*
4. *$K_i\varphi \to B_i\varphi$, $B_i\varphi \to K_iB_i\varphi$, and $K_i\varphi \to \Box_i\varphi$.*

Proof sketch. (1) is immediate from $\sim_i$ being an equivalence relation. (2): seriality follows from Proposition 2.5 applied to $\sim_i^u$, which is non-empty since $u \sim_i u$; introspection follows because $\to_i^u$ depends only on the class of u, and all worlds in a class have the same class. Failure of T is witnessed by any model in which $u \notin \to_i^u$. (3): reflexivity of R_i places u in its own quantification domain. (4): the inclusions of Theorem 3.3, plus the class-invariance of $\to_i^u$ for the second conjunct. $\Box$

Remark 3.5. The asymmetry in (2)—consistency but not factivity—is the formal content of the informal claim that an agent may be *wrong but not incoherent.* $B_i\varphi \wedge \neg\varphi$ is satisfiable; $B_i\varphi \wedge B_i\neg\varphi$ is not.

3.4 Derived attitudes

The implemented core has exactly the eight cases of Definition 3.2. Every further attitude in Table 1 is defined by translation and eliminated before evaluation.

The "whether" operators are worth singling out. $Kw_a\varphi$ is not $K_a\varphi$ and not its negation: it is a *disjunction* of two knowledge claims, asserting that the agent has settled the question without saying which way. Questions about ignorance are its negation. That these require a disjunction is what makes them operators rather than abbreviations a user can be expected to expand by hand, and Section 6 shows that the same disjunctive structure is what lets the enumeration procedure reach them.

3.5 Safe belief

Safe belief occupies the gap in Theorem 3.3, and because it is measured from the actual world its behaviour is asymmetric in a way worth developing.

Example 3.6 (Three positions on one question)**.** Consider $\mathcal{P} = \{up\}$ and three agents, none of whom knows the answer:

```
initially {
    up                              // the server really is up
    ?[ada] up    B[ada] up          // ada cannot tell, but leans the right way
    ?[ben] up    B[ben] !up         // ben cannot tell, and leans the wrong way
    ?[cleo] up                      // cleo has no leaning either way
}
```

Then $B_{ada}\,up$ and $B_{ben}\,\neg up$ both hold, so belief cannot separate the first two agents. Safe belief separates all three: $\Box_{ada}\,up$ holds, $\Box_{ben}\,\neg up$ does not (indeed cannot, by factivity), and $\Box_{cleo}\,up$ does not, because both worlds are maximal for *cleo* and so both lie above the actual one.

Proposition 3.7 (Defeasibility characterisation)**.** *$\Box_i\varphi$ holds at $\langle M, u\rangle$ iff φ is believed and remains believed after learning any ψ true at u; equivalently, iff $B_i^{\psi}\varphi$ holds for every ψ with $\langle M, u\rangle \models \psi$.*

This is the characterisation of Baltag and Smets (2008), and it is what makes $\Box_i$ more than a technical interpolant: a safe belief is one that *no true information can overturn.* The qualification matters—a safe belief may still be overturned by a falsehood, and Section 3.5.2 exhibits a case.

3.5.1 Introspection

Because $\Box_i$ is anchored to the actual world, and the actual world is precisely what an agent cannot locate, safe belief is not introspectible in the way knowledge and belief are.

Theorem 3.8 (Introspection collapse)**.** *On every plausibility model:*

$$K_i\Box_i\varphi \;\equiv\; K_i\varphi, \qquad B_i\Box_i\varphi \;\equiv\; B_i\varphi, \qquad \Box_i\Box_i\varphi \;\equiv\; \Box_i\varphi.$$

Proof. First. ($\Leftarrow$) If φ holds throughout $\sim_i^u$ then in particular it holds throughout every up-set contained in that class, so $\Box_i\varphi$ holds at every $v \in \sim_i^u$. ($\Rightarrow$) Suppose $K_i\Box_i\varphi$ at u, and let v_0 be a minimal element of $\sim_i^u$, which exists because the class is finite and, by local connectedness, totally preordered. Then $\{w : v_0 R_i w\} = \sim_i^u$, and $\Box_i\varphi$ at v_0 gives φ throughout the class, i.e. $K_i\varphi$ at u.

Second. Let $v \in \to_i^u$. Since v is maximal in its class, $\{w : vR_iw\} = \to_i^u$. Hence $\Box_i\varphi$ at v iff φ holds throughout $\to_i^u$, which is $B_i\varphi$ at u; and this is independent of the choice of v.

Third. Transitivity of R_i gives $\Box_i\varphi \to \Box_i\Box_i\varphi$; reflexivity gives the converse. $\Box$

Corollary 3.9 (Overconfidence)**.** *Every agent believes each of its beliefs to be safe: $B_i\varphi \to B_i\Box_i\varphi$ is valid, including when $\Box_i\varphi$ is false.*

Corollary 3.9 is not a defect of the semantics but a consequence of belief being evaluated at the maxima: from the top of one's own ordering, every belief looks unshakeable. In Example 3.6, $B_{ben}\Box_{ben}\neg up$ holds while $\Box_{ben}\neg up$ does not; *ben* takes his belief to be immune to revision at the moment before one true announcement overturns it.

The practical consequence for modelling is that $\Box_i$ is an *external* operator. It measures the fit between an agent's ordering and the actual world; the agent has no access to the second argument, and its self-report is uninformative because it is unconditionally affirmative. An analyst may ask $\Box_i\varphi$ of a state; an agent may not ask it of itself.

Route	Instance	K_i	$\Box_i$	B_i
Sensing	the agent looks	✓	✓	✓
Truthful announcement	the agent is told, and it is true	—	✓	✓
Lie	the agent is told a falsehood	—	—	✓

Table 2: Acquisition of the three attitudes toward the announced proposition. The middle row is the case that motivates the operator: a truthful announcement upgrades a previously mistaken belief to a safe one without conferring knowledge.

3.5.2 How safe belief is acquired

An agent safely believes φ exactly when the actual world is ranked at least as high as every world where φ fails. Table 2 records the three dynamic routes, anticipating the action layer of Section 5.

Two observations. First, the middle row describes a state with no compact name in ordinary usage: the agent is now right, no true information can dislodge it, and it still cannot rule out the alternative. Second, the bottom row cannot come out otherwise, since $\Box_i$ is factive—but a lie can nevertheless *destroy* a safe belief, by reordering the worlds of an agent that previously held one. Safety is immunity to truth, not permanence.

4 The surface language

A delhi domain is a text file with a fixed section structure. We give the grammar, then treat the sections whose design is not immediate.

4.1 Structure

```
types       { Actor - Object }          // type hierarchy
objects     { alice, bob - Actor }      // typed constants
agents      { alice, bob }              // which objects have minds
props       { h, d }                    // propositional atoms
constants   { adjacent(hall, study) }   // static, world-independent
define      { ... }                     // named formulas, expanded away
rules       { ... }                     // Horn clauses over constants
initially   { ... }                     // declarative initial state
state       { ... }                     //    ...or an explicit model
goal        { phi }                     // optional
invariants  { phi ... }                 // must hold in every state
actions     { ... }                     // action theory
```

Listing 1: Overall file structure. `initially` or `state` is required, as is `actions`, which may be empty.

4.2 Declarative initial states

The design decision most visible to a user is that initial states are given as *constraints* rather than as structures. An `initially` block lists facts and attitudes; delhi constructs a model satisfying them and then verifies the model against every line written.

```
initially {
    h                       // a fact about the actual world
```

```
    ?[carol] h            // carol cannot tell whether h
    B[carol] h            // but she correctly leans that way
}
```

Anything not mentioned is known by every agent, so the block above yields a two-world model in which *carol* is uncertain and the others are not. The alternative form, `state`, writes the model out directly and is what the tool prints when asked to display one:

```
state {
  *w1 <- { h }          // ‘*‘ marks the designated world
   w0 <- { }
  carol: w0 < w1        // w1 is the more plausible
}
```

Here `<` and `<=` point toward the more plausible world and relates two worlds in both directions. Agents with no line distinguish every world, which is to say they know everything—the degenerate case of Definition 2.1 in which R_i is the identity.

4.3 Definitions, rules and invariants

Three optional sections address needs that arise in practice and that the underlying semantics does not itself provide.

Definitions. Named formulas, parameterised over objects, expanded before lowering so that the semantics never encounters the name:

```
define {
    blocked(?r)       = !lit(?r) | locked(?r)
    can_enter(?w, ?r) = !blocked(?r) & K[?w] !blocked(?r)
}
```

Parameters may occupy agent positions, as `?w` does above. Cycles are rejected when the definition table is built rather than caught by a depth limit at expansion time. Two restrictions are deliberate: a defined name may not appear as the target of `causes` or as a world fact, since both require an atom the semantics can *set*; and parameters range over objects, not formulas, so second-order macros such as `define f(?p) = K[a] ?p` are not admitted.

Rules. Horn clauses over the static constant table, saturated to a least fixpoint at parse time:

```
constants { !adjacent(Room, Room)  adjacent(hall, study)  adjacent(study, attic) }
rules {
    reach(?x, ?y) :- adjacent(?x, ?y)
    reach(?x, ?z) :- adjacent(?x, ?y), reach(?y, ?z)
}
```

Derived predicates never become propositions, and therefore consume no bit in any world; `reach(hall, attic)` folds to $\top$ exactly as a declared constant would, and ground actions whose guards fold to $\perp$ are discarded before construction.

The restriction to *constants* is the substantive one. The fixpoint is computed once, which is sound only because the extension is world-independent. A rule over a fluent would have an extension varying per world and per action, so evaluating it would require either a fixpoint per world at query time or the maintenance of derived atoms through product update—the frame problem in a new place. delhi rejects such rules with a diagnostic rather than supporting them partially. Bodies carry no negation, keeping the program monotone so that the least fixpoint exists, and every head variable must occur in the body, so that no head asserts more than the body justifies.

Invariants. Constraints required to hold in *every* state rather than only the first:

```
invariants { !((B[a] p & B[b] !p) | (B[a] !p & B[b] p)) }
```

An `initially` entry that drives no construction is already an assertion about the start; an invariant is the same assertion made about the whole run, which is usually what a domain constraint is intended to mean. Invariants are checked when the initial state is built and after every action.

5 Dynamics

An action changes a state by product update with an event model. The distinguishing feature of the action-language approach, inherited from mB and $m\mathcal{A}^*$, is that the modeller does not construct event models: they declare *what changes* and *who observes*, and the event model is derived.

5.1 Action theories

Definition 5.1 (Action theory)**.** An action theory T for action a is a set of statements of the following forms, with the indicated formula typing:

#	Form	Typing	Role
1	a **requires** φ	$\varphi \in \mathcal{L}^{\mathcal{P}}$	precondition
2	a **causes** $l_0, \ldots, l_n$ **if** φ	l_j literals, $\varphi \in \mathcal{L}^{\mathcal{P}}$	ontic effect
3	a **determines** φ	$\varphi \in \mathcal{L}^{\mathcal{P}}$	sensing
4	a **announces** ψ	$\psi \in \mathcal{L}^{\mathcal{P}}_{\mathcal{G}}$	announcement
5	i **observes** a **if** φ	$\varphi \in \mathcal{L}^{\mathcal{P}}$	full observer
6	i **aware_of** a **if** φ	$\varphi \in \mathcal{L}^{\mathcal{P}}$	partial observer

An action theory is *well-formed* when it contains exactly one statement of form 2, 3 or 4; at most one of form 1; no `causes` list containing both p and $\neg p$; and, for every state, no agent satisfying both an `observes` and an `aware_of` condition.

Remark 5.2 (Typing is load-bearing)**.** Form 4 admits a *modal* ψ, which need not be true. This is what makes announcements capable of being lies, and of being about beliefs rather than facts. It also propagates: announcement event preconditions are $a_{pre} \wedge \psi$ and are therefore modal, so precondition evaluation requires full modal entailment against the pre-update model. Edge conditions, by contrast, remain propositional, being Boolean combinations of observability conditions—a distinction worth preserving in an implementation, since edge conditions are evaluated at two worlds per edge per agent and dominate the cost of update.

Remark 5.3 (Preconditions: a divergence in the sources)**.** The source documents give three incompatible readings of multiple form-1 statements: disjunction (Buckingham, 2021, eq. 5.1), at most one statement, and—in the reference implementation's corpus—conjunction. Under the disjunctive reading, *adding* a precondition makes an action *more* applicable, contradicting both the ordinary meaning of the word and every domain file in that corpus. delhi adopts the at-most-one rule: a single `pre` clause per action, conjunction written explicitly, and $a_{pre} = \top$ in its absence. A second `pre` clause is a diagnostic rather than a silent combination.

Position	**Declaration**	**Learns**	**Labels**
Full observer	`i observes a`	what happened, outcome included	$PN(i) = \bot$, $N(i) = \bot$
Partial observer	`i aware a`	that it happened, not how	$PN(i) = \top$, $N(i) = \bot$
Oblivious	neither	nothing	$PN(i) = \top$, $N(i) = \top$

Table 3: The three observer positions, with the edge labels each induces (for unconditional declarations). *PN* governs whether the agent can distinguish the outcome events; *N* governs whether it can distinguish them from the null event.

Kind	**Construction**
Ontic	$E = \{e^c, e^\top\}$; $Q(e^c, e^\top) = N$; $pre(e^c) = a_{pre}$, $pre(e^\top) = \top$; $add(e^c) = P^+$, $del(e^c) = P^-$; $\Gamma = \{e^c\}$. With conditional effects, e^c splits into one event per realisable outcome with mutually exclusive preconditions.
Announcement	$E = \{e^\varphi, e^{\neg\varphi}, e^\top\}$; $Q(e^\varphi, e^{\neg\varphi}) = PN$, $Q(e^{\neg\varphi}, e^\varphi) = FPN$, $Q(e^\varphi, e^\top) = Q(e^{\neg\varphi}, e^\top) = N$; $pre(e^\varphi) = a_{pre} \wedge \varphi$, $pre(e^{\neg\varphi}) = a_{pre} \wedge \neg\varphi$, $pre(e^\top) = \top$; $add = del = \emptyset$; $\Gamma = \{e^\varphi, e^{\neg\varphi}\}$.
Sensing	As announcement, but $Q(e^{\neg\varphi}, e^\varphi) = PN$ rather than *FPN*.

Table 4: The three event-model constructions. P^+ and P^- are the positive and negative literals of the `causes` list.

5.2 Observability

Each agent occupies exactly one of three positions with respect to an action. The three are distinguished by two edge labels, defined from the observability conditions of forms 5 and 6:

$$FPN(i) := \top \tag{13}$$

$$PN(i) := \neg \bigvee\nolimits_{\texttt{observes}\ \varphi} \varphi \tag{14}$$

$$N(i) := \neg \left(\bigvee\nolimits_{\texttt{observes}\ \varphi} \varphi \ \vee\ \bigvee\nolimits_{\texttt{aware\_of}\ \varphi} \varphi\right) \tag{15}$$

Table 3 makes precise what is otherwise easy to state loosely. A partial observer retains the edge between the two outcome events—so the outcomes remain indistinguishable to it—and loses the edge to the null event, so it does rule out that nothing happened. That combination is exactly the informal reading "knows the action occurred, not how it turned out". The consequence is second-order: if j senses φ and i is a partial observer, i comes to know that j knows *whether* φ, without learning φ.

Both declarations admit a condition, evaluated in the state at the time of the action. This is what Buckingham (2021) calls *local dynamic observability*, and it is the mechanism by which an agent's observer position may change during a trace—for instance, an agent that is a partial observer only while not distracted.

5.3 Event models and the three constructions

Definition 5.4 (Action plausibility model)**.** An *action plausibility model* is $\langle E, Q, pre, add, del, \Gamma\rangle$ with E a finite set of events, $Q : E \times E \to (\mathcal{G} \to \mathcal{L}^{\mathcal{P}})$ the edge conditions, $pre : E \to \mathcal{L}^{\mathcal{P}}_{\mathcal{G}}$, $add, del : E \to 2^{\mathcal{P}}$, and $\Gamma \subseteq E$ designated. Every event carries a reflexive *FPN* edge for every agent; unlisted edges are $\bot$.

The single-cell difference between announcement and sensing in Table 4 carries the whole distinction between being told and finding out. With $FPN = \top$ on the backward edge, even a full observer of an announcement retains an edge from $e^{\neg\varphi}$ to e^φ, and so cannot rule out that what was said was false; the result is belief. With *PN* there, a full observer distinguishes the two events and comes to *know* whether φ.

5.4 Product update

Definition 5.5 (State transition)**.** *Write $e \longrightarrow^{iuv} f$ for $\langle M, u\rangle \models Q(e, f)(i)$ and $\langle M, v\rangle \models Q(e, f)(i)$. The updated model $M' = \langle W', R', V'\rangle$ is*

$$W' = \{\langle u, e\rangle \mid u \in W,\ e \in E,\ \langle M, u\rangle \models pre(e)\} \tag{16}$$

$$R'(\langle u, e\rangle, \langle v, f\rangle)(i) = \begin{cases} (e \longrightarrow^{iuv} f) \wedge \neg(f \longrightarrow^{iuv} e) \wedge\ u \sim_i v, & \text{or} \\ (e \longrightarrow^{iuv} f) \wedge (f \longrightarrow^{iuv} e) \wedge\ uR_i v \end{cases} \tag{17}$$

$$V'(\langle u, e\rangle) = (V(u) \cup add(e)) \setminus del(e) \tag{18}$$

with designated world $\langle d, e\rangle$ for the unique $e \in \Gamma$ whose precondition holds at d.

This is Baltag–Smets action-priority update: event plausibility overrides state plausibility, so incoming information takes precedence over prior belief except where it contradicts prior knowledge.

Remark 5.6 (A divergence, and why it is decisive)**.** An alternative rule appears in the earlier literature, differing only when $f \longrightarrow^{iuv} e$ holds and $e \longrightarrow^{iuv} f$ does not—that is, when e is strictly preferred. Definition 5.5 creates no edge, preserving a strict preference; the alternative creates one conditioned on $uR_i v$, which can render the two worlds equiplausible.

This is not a corner case requiring construction. In the standard *Coin Lie* scenario, at the lied-to agent, from $\langle w_1, e^{\psi}\rangle$ to $\langle w_0, e^{\neg\psi}\rangle$: the forward label is *PN*, which for a full observer is $\neg\top = \bot$, and the backward label is $\mathit{FPN} = \top$. Under Definition 5.5 no edge is created, the agent retains a strict preference for the announced world, and the lie lands. Under the alternative the worlds become equiplausible and the lie fails to land at all. Since the scenario is the one the source uses to *demonstrate* second-order false belief, the alternative rule breaks the example it was introduced to explain. delhi implements Definition 5.5; both are implemented behind a flag and the divergence is pinned by a regression test.

6 The query language

6.1 Checking

A query is a formula of $\mathcal{L}_G^{\mathcal{P}}$ extended with a hole. Checking is satisfaction:

$$\text{eval}(s, \varphi) \ = \ s \models \varphi, \qquad \varphi \in \mathcal{L}_G^{\mathcal{P}},$$

total and decidable, since $\models$ is evaluated over a finite model with memoisation.

```
query    ::= formula
formula  ::= atom | 'true' | 'false' | '_'
           | '!' formula
           | formula ('&' | '|' | '->') formula
           | modality '[' agents ']' formula
           | 'B' '^' formula '[' agents ']' formula
           | '(' formula ')'
modality ::= 'K' | 'B' | '[]' | 'C' | "K'" | "B'" | "S'"
           | 'Kw' | 'Bw' | '?' | '??'
agents   ::= '*' | name (',' name)*
```

Listing 2: Query grammar. Precedence, loosest to tightest: `->` (right-associative), `|`, `&`, `!`, modality.

The query language is *exactly* the formula language plus _: every operator of Definition 3.2 and every derived attitude of Table 1 is available, closed under the Boolean connectives at arbitrary depth. Goals, invariants and interactive queries share one parser and one lowering path, so nothing expressible as a goal is inexpressible as a query.

Remark 6.1 (The hole is a token)**.** `_` lexes as a distinct token; `_x` and `at_park` remain ordinary identifiers. This is load-bearing rather than cosmetic. Filling by textual substitution tears any identifier containing an underscore, turning `_ & at_park` into a malformed formula. Filling is a tree substitution, which additionally means a filled pattern requires no defensive parentheses: $\neg\_$ negates whatever fills it regardless of the filler's precedence.

6.2 Enumeration

Checking answers a question the analyst already knows to ask. The complementary operation enumerates the formulas of a given *shape* that hold.

Definition 6.2 (Ask)**.** For a state s, a pattern π containing at least one hole, and a depth d,

$$\text{ask}(s, \pi, d) \;=\; \{\, \pi[\varphi/\_] \mid \varphi \in ML_d,\; s \models \pi[\varphi/\_] \,\},$$

where every hole in π receives the *same* φ and ML_d is the set of *modal literals* of depth at most d:

$$Lit = \{\, p,\ \neg p \ \mid\ p \in \mathcal{P} \,\} \tag{19}$$

$$ML_0 = Lit \tag{20}$$

$$ML_k = ML_{k-1} \cup \{\, O_i\varphi \ \mid\ O \in \{K, B\},\ i \in \mathcal{G},\ \varphi \in ML_{k-1} \setminus ML_{k-2} \,\}. \tag{21}$$

Hence $|ML_d| = \sum_{k \le d} (2|\mathcal{G}|)^k \cdot 2|\mathcal{P}|$.

Candidates are enumerated shallowest-first, so a truncated answer is a prefix rather than a sample, and truncation is reported: a partial answer must never be readable as a complete one.

Remark 6.3 (Why the domain is restricted)**.** The set $\{\varphi : s \models B_a\varphi\}$ is infinite—conjunction alone generates without bound—so "everything a believes" is not enumerable, and no implementation choice changes this. ML_d is the restriction adopted by Muise et al. (2015) for depth-bounded epistemic planning; here it serves the same purpose, being finite with a size determined by the signature and the depth.

Remark 6.4 (What the restriction costs)**.** The restriction concerns Boolean structure *inside* a modality only; structure outside is untouched, because the pattern is an arbitrary formula. $Kw_a\varphi$ is by definition $K_a\varphi \vee K_a\neg\varphi$, and enumeration reaches it either through the sugar or written out as `K[a] _ | K[a] !_`, the two holes taking one filler.

Inside a modality, conjunctive candidates would be redundant, since K and B are normal and $K_a(\varphi \wedge \chi) \equiv K_a\varphi \wedge K_a\chi$. Disjunctive candidates would *not* be: $B_a(p \vee q)$ may hold when neither B_ap nor B_aq does, which is the state of knowing that one of two things holds without knowing which. That is the one genuinely missing class, and the natural direction in which to widen enumeration.

Table 5 illustrates why uniform filling of repeated holes is the design's load-bearing choice. The false-belief pattern $B_a\_ \wedge \neg\_$ asks for exactly the conjunction that Section 1 identified as the object of the Sally–Anne task, and it finds instances the analyst had no reason to anticipate.

7 Implementation

delhi is implemented in Rust in six crates: a hash-consed formula store, a set of shared traits, the semantic core, the front end, the command-line tool, and an optional browser interface. Every crate but the last has no external dependencies.

Question	Pattern
What does a believe?	`B[alice] _`
What can a not settle?	`?[alice] _`
What does a think c believes?	`B[alice] B[carol] _`
What does a believe that is false?	`B[alice] _ & !_`
Where do a and c disagree?	`B[alice] _ & B[carol] !_`
What does a know that c does not?	`K[alice] _ & !K[carol] _`
What does c believe without knowing?	`B[carol] _ & !K[carol] _`

Table 5: Enumeration patterns. The repeated hole is what makes the last four expressible; each occurrence receives the same filler.

7.1 Representation

Three decisions account for most of the performance behaviour reported in Section 8.

Hash-consed formulas. Formulas are interned in a store that guarantees structural identity implies pointer identity. Subformula equality is therefore an integer comparison, and memoisation tables can be keyed on formula identifiers.

Bitset models. Valuations are bitsets over $\mathcal{P}$ and relations are bitsets over $W \times W$ per agent. The dominant cost of update is the edge-condition evaluation of Definition 5.5, performed at two worlds per edge per agent; representing the relation densely makes that loop cache-friendly at the cost of $|\mathcal{G}| \cdot |W|^2$ bits.

Memoised entailment. Satisfaction is computed by memoised recursion keyed on (world, formula). Common knowledge is evaluated as a reachability closure rather than by unfolding.

7.2 Bisimulation contraction

Product update multiplies: each action crosses every world with every distinguishable event, so iterated update grows exponentially unless the model is quotiented. delhi implements two notions.

Definition 7.1 (Two bisimulations)**.** $\underline{\leftrightarrow}_R$ is the standard relational bisimulation refining on R_i and R_i^{-1} in both directions. $\underline{\leftrightarrow}_D$ refines on the *derived* relations $\sim_i$, $\to_i$ and R_i that the satisfaction clauses of Definition 3.2 actually use.

Proposition 7.2. *$\underline{\leftrightarrow}_R$ is sound—bisimilar states satisfy the same formulas of $\mathcal{L}_{\mathcal{G}}^{\mathcal{P}}$—and is a congruence for product update. It is not complete: it may distinguish states that no formula separates.*

The incompleteness has an identifiable cause. $\underline{\leftrightarrow}_R$ refines against R_i^{-1}, whereas no operator of Definition 3.2 quantifies over the inverse relation; the extra refinement is therefore invisible to the language. $\underline{\leftrightarrow}_D$ removes it and coincides exactly with modal equivalence, but its status as a congruence for update is open, so delhi uses $\underline{\leftrightarrow}_R$ wherever a contracted state is fed back into an update and offers $\underline{\leftrightarrow}_D$ for measurement. In practice the gap is small: on random models we measure a 5–10% difference in resulting world counts, and on every domain in our example corpus the two produce *identical* counts.

Cycle	No contraction		$\underline{\leftrightarrow}_R$		$\underline{\leftrightarrow}_D$	
	worlds	cumulative	worlds	cumulative	worlds	cumulative
0	2	—	2	—	2	—
1	16	72.9 µs	14	205.7 µs	14	168.7 µs
2	128	2.2 ms	16	797.7 µs	16	903.3 µs
3	1024	133.6 ms	16	1.4 ms	16	1.7 ms
4	8192	9.47 s	16	2.0 ms	16	2.5 ms

Table 6: *Coin Lie*, one cycle being its three actions. Uncontracted growth is multiplicative in the number of distinguishable events; cost grows faster still, because the relation is $|\mathcal{G}| \cdot |W|^2$ bits and almost every operation traverses it.

Cycle	Worlds ($\underline{\leftrightarrow}_R$)	Cumulative
0	8	—
1	36	528.1 µs
2	92	—
3	188	—
4	588	—
5	2460	—
6	7276	—

Table 7: *Grapevine* under $\underline{\leftrightarrow}_R$ contraction. The cycle—an agent leaves, a secret is shared, the agent returns—creates a genuinely new distinction each time round, because each repetition adds a layer of who-was-present-for-what. Contraction cannot merge worlds that are really different.

8 Evaluation

We report model growth and elapsed time under iterated update, with and without contraction. Measurements are single-threaded on a commodity laptop; the interest is in the growth curves rather than the constants.

8.1 Growth without contraction

Table 6 shows the uncontracted trajectory reaching 8192 worlds and 9.47 seconds by the fourth cycle—already unusable—while both contracted trajectories settle at 16 worlds and roughly 0.6 ms per cycle, making total cost linear in the number of actions rather than exponential. *Muddy Children* behaves the same way, settling at 32 worlds where the uncontracted run reaches 8192 and 8.8 seconds.

8.2 A fixed point is not the general case

It is tempting to conclude that contraction bounds model size. It does not.

Table 7 records a domain in which contraction does not reach a fixed point. This is not a deficiency of the quotient: the worlds genuinely differ, and any sound contraction must keep them apart. The practical consequence is that contraction is necessary but not sufficient, and that a forward-search planner built on this semantics will require depth bounding or an additional abstraction, not merely a canonical state key.

8.3 Correctness

The implementation carries 287 tests. These include the frame properties of Theorem 3.4 as property tests over generated models, the identity $B_i^{\top}\varphi \equiv B_i\varphi$, the divergence of Remark 5.6 as a pinned regression, and, for each example domain, an assertion of the headline claim the domain exists to demonstrate—so that the documentation and the implementation cannot drift apart silently.

9 Related work

Epistemic planning has two traditions in tension. One begins from *expressiveness*: dynamic epistemic logic (Baltag et al., 1998; van Ditmarsch et al., 2007) will represent essentially anything, at the cost of requiring the modeller to hand-build event models and of undecidable plan existence (Bolander and Andersen, 2011). The other begins from *tractability*, restricting the representation until established planning machinery applies. Action languages occupy the middle: they retain a semantics grounded in DEL while deriving event models from declared observability. mB^+ is in this third group, and specifically in the branch that replaced knowledge with belief.

	Belief ≠ knowledge	**Revision on contradiction**	**2nd-order false belief**	**False belief about observers**	B_i^{ψ} / $\Box_i$
DEL	yes	via update rules	yes	yes	in extensions
Baltag & Smets	yes	yes (its origin)	yes	—	**yes**
$m\mathcal{A}^*$	limited	collapses all uncertainty	no	no	no
$m\mathcal{A}^*$ + h.o. obs.	limited	as $m\mathcal{A}^*$	yes	yes	no
mB	yes	yes, preserving other uncertainty	yes	yes	models only
$\mathbf{mB^+}$ **/ delhi**	yes	yes	yes	yes	**yes, in the language**
EFP / EFP 2.0	knowledge-oriented	—	—	—	no
PDKB / RP-MEP	yes	bounded	to the depth bound	no	no

Table 8: Expressiveness. "Limited" for $m\mathcal{A}^*$ records a deliberate trade of expressiveness for planning performance, not a shortcoming.

9.1 What is transcribed and what is new

Intellectual honesty requires separating the two. Transcribed from Buckingham (2021) and the associated papers: the plausibility-model semantics, the six statement forms of Definition 5.1, the observability model including local dynamic observability, the three constructions of Table 4, and the product update of Definition 5.5. Taken from Baltag and Smets (2006, 2008): the operators $\Box_i$ and B_i^{ψ} and the action-priority discipline. The name mB^+ is ours and appears in no source.

New here, to the best of our knowledge:

1. Lifting $\Box_i$ and B_i^{ψ} into the object language of an action formalism, making them queryable properties of a planning state rather than metatheoretic notions (Section 3), together with the introspection analysis of Theorem 3.8 and Corollary 3.9.

2. The enumeration procedure of Definition 6.2: patterns with repeated holes filled uniformly, over the modal-literal domain of Muise et al. (2015) used as a *search space* rather than as

	Event models	State representation	Planner
DEL	hand-built per problem	Kripke models	none; existence undecidable
Baltag & Smets	action-priority update	plausibility models	none (a logic)
$m\mathcal{A}^*$	derived from observability	Kripke models	yes (ASP or forward search)
mB	derived from observability	plausibility models	yes
delhi	derived from observability	plausibility models, bitset-backed	**not yet**
EFP 2.0	derived	possibilities / Kripke	yes, heavily optimised
PDKB	—	depth-bounded knowledge bases	yes, compiled to classical

Table 9: Machinery. The planner column is where delhi is behind: EFP 2.0 and PDKB are mature planning systems, whereas delhi is a model checker with the components for a planner present but unused.

a state representation.

3. Declarative initial states (Section 4.2) with post-construction verification, and the auxiliary language layers—definitions, Horn rules over constants, invariants.

4. Conditional ontic effects combined with list-valued `causes`, which the two source presentations offer only separately.

5. The measurement of Section 8, in particular the demonstration that contraction reaches a fixed point on some domains and provably not on others, and the diagnosis of $\underline{\leftrightarrow}_R$'s incompleteness as refinement against an inverse relation no operator uses.

10 Conclusion

We have given the syntax and semantics of mB^+ and of the delhi language that realises it, covering the static logic, the surface language, the action layer, and a query language with an enumeration procedure. Three findings seem to us to have interest beyond the implementation.

First, safe belief behaves quite differently from knowledge and belief with respect to introspection: $K_i\Box_i\varphi \equiv K_i\varphi$ and $B_i\Box_i\varphi \equiv B_i\varphi$, so an agent can never certify a safe belief short of knowledge and unconditionally believes its beliefs to be safe. Safe belief is thus an analyst's operator, not an agent's.

Second, the choice of product-update rule is not a matter of taste. The two rules in the literature diverge precisely on the configuration that the standard demonstration scenario exercises, and one of them prevents that scenario's conclusion from arising at all.

Third, bisimulation contraction is necessary but not sufficient for tractable iterated update. It converts exponential growth into a fixed point on several standard domains and demonstrably does not on others, which bears directly on the design of any planner built over this semantics.

Future work. The obvious next component is a forward-search planner over the contracted state space; the canonical state keys and the contraction it requires are already present. Two open questions bear on it: whether $\underline{\leftrightarrow}_D$ is a congruence for product update, which would allow the tighter quotient to be used inside search; and how to bound or abstract domains of the *Grapevine* kind, where sound contraction cannot prevent growth. On the language side, the natural extension is to admit formula-valued postconditions in the style of Andersen–Bolander–Jensen, which subsume conditional effects, together with a determination of what observers should learn from them.

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